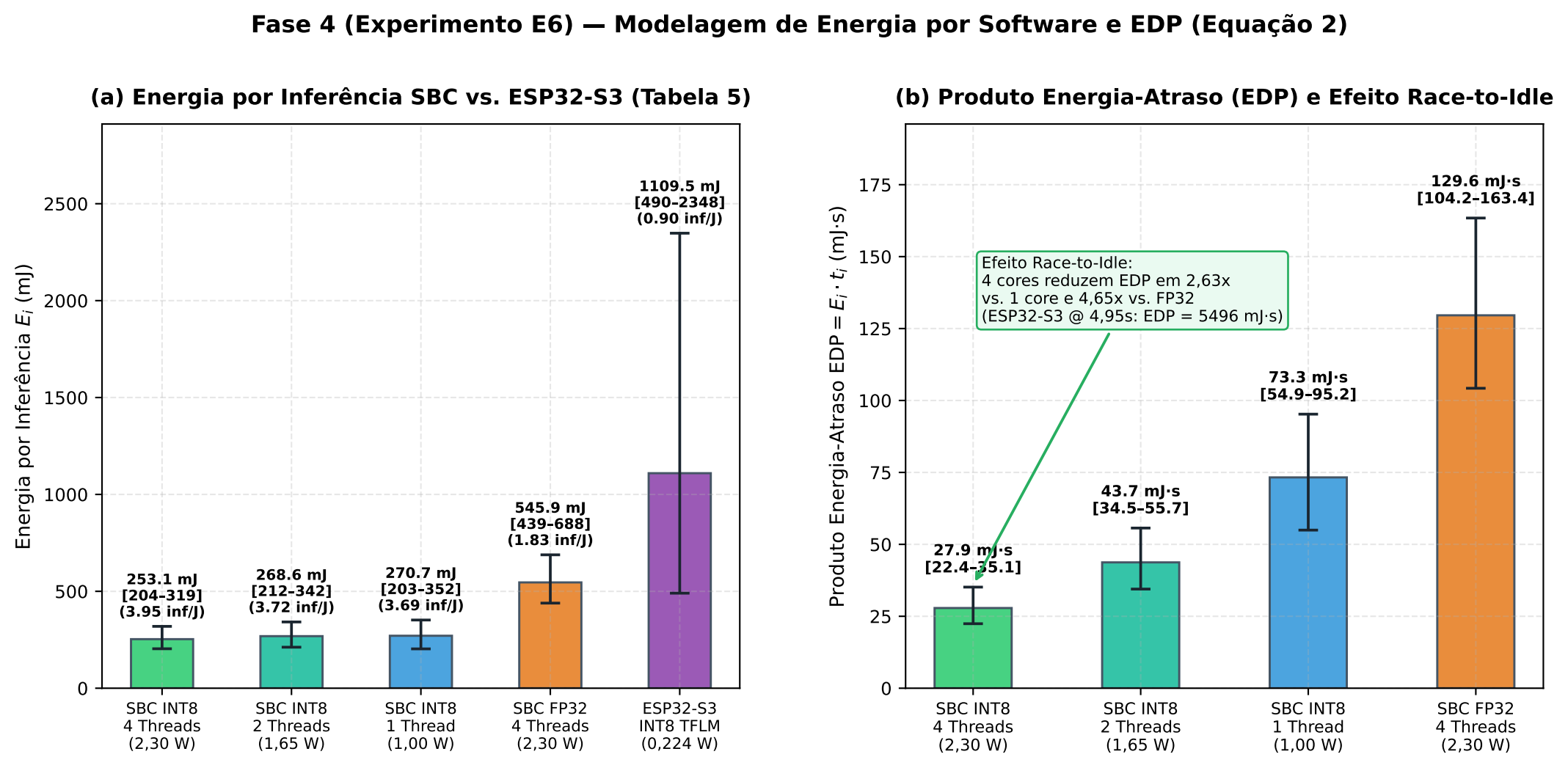
Fase 4 (Experimento E6) — Modelagem de Energia por Software e EDP (Equação 2)
(a) Energia por Inferência SBC vs. ESP32-S3 (Tabela 5)
Energia por Inferência E_i (mJ)
0
500
1000
1500
2000
2500
253.1 mJ
[204–319]
(3.95 inf/J)
268.6 mJ
[212–342]
(3.72 inf/J)
270.7 mJ
[203–352]
(3.69 inf/J)
545.9 mJ
[439–688]
(1.83 inf/J)
1109.5 mJ
[490–2348]
(0.90 inf/J)
SBC INT8
4 Threads
(2,30 W)
SBC INT8
2 Threads
(1,65 W)
SBC INT8
1 Thread
(1,00 W)
SBC FP32
4 Threads
(2,30 W)
ESP32-S3
INT8 TFLM
(0,224 W)
(b) Produto Energia-Atraso (EDP) e Efeito Race-to-Idle
Produto Energia-Atraso EDP = $E_i \cdot t_i$ (mJ·s)
0
25
50
75
100
125
150
175
Efeito Race-to-Idle:
4 cores reduzem EDP em 2,63x
vs. 1 core e 4,65x vs. FP32
(ESP32-S3 @ 4,95s: EDP = 5496 mJ·s)
27.9 mJ·s
[22.4–35.1]
43.7 mJ·s
[34.5–55.7]
73.3 mJ·s
[54.9–95.2]
129.6 mJ·s
[104.2–163.4]
SBC INT8
4 Threads
(2,30 W)
SBC INT8
2 Threads
(1,65 W)
SBC INT8
1 Thread
(1,00 W)
SBC FP32
4 Threads
(2,30 W)